
SOUTHEAST ASIA ENGINEERING DESIGN COMPETITION 2026

BinSight

General-waste sensing, recycling vision
and route simulation

Team: MON BLUE

Three general-waste bins | Vision-enabled recycling return | Decision hub | Route simulation

Engineering Proposal

Problem Statement

Collection crews still follow fixed routes, causing them to empty underfilled bins while high-use sites overflow. This wastes fuel, increases collection emissions, and leaves operators without real-time visibility into which bins need urgent service.

Proposed Solution

BinSight has two hardware paths. Focus A instruments three general-waste bins using one Teensy 4.1 and FreeRTOS for scheduled ultrasonic/load-cell sensing, watchdog recovery, fault flags and fill telemetry through an ESP32-C3 Wi-Fi relay. Focus C's laptop hub logs readings, estimates time-to-overflow with a tree-based model, ranks pickups and compares fixed schedules with priority routes. Focus B adds vision only to the recycling-return station: an OV5647 camera feeds Grove Vision AI V2 for local classification. Focus D's dedicated ESP32-C3 relays class, confidence and timing, manages the QR session and commands chute/servo feedback; it does not run the model. Accepted plastic, metal and glass items are logged for a simulated refund, while non-recyclables are rejected.

Simulation / Proposed Targets

The simulation uses prototype logs in a 30-day district model. It compares fixed collection with the priority route and checks for fewer overflows, fewer wasted trips, shorter route distance, lower fuel and CO₂, cleaner return-station streams, and lower sensing energy within the USD150/SGD200 build.